

Functional Annotation Report: *mqo3* (PP_2925, UniProt Q88IS4) — Probable Malate:Quinone Oxidoreductase 3 of *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

mqo3 (locus tag **PP_2925**; KEGG symbol *mqo-III*; UniProt **Q88IS4**) of *Pseudomonas putida* KT2440 encodes one of three paralogous **malate:quinone oxidoreductases** (MQO; EC 1.1.5.4). Its primary, and essentially sole, catalytic function is the **FAD-dependent oxidation of (S)-malate (L-malate) to oxaloacetate, transferring the two electrons directly to a membrane quinone** rather than to NAD⁺:

(S)-malate + a quinone → oxaloacetate + a quinol (Rhea:46012; EC 1.1.5.4)

This is the malate → oxaloacetate step of the **tricarboxylic acid (TCA/citric acid) cycle**, performed via the "quinone route" (UniProt/HAMAP pathway: *TCA cycle; oxaloacetate from (S)-malate (quinone route): step 1/1*). Because the electron acceptor is a lipophilic quinone of the respiratory chain, MQO3 acts as a **membrane-associated peripheral flavoprotein that couples the TCA cycle directly to aerobic respiration**, in the same conceptual manner that succinate dehydrogenase (complex II) does. The enzyme carries out its function on the **cytoplasmic face of the inner (cytoplasmic) membrane**, where it can reduce the membrane ubiquinone pool. Its substrate specificity is for L-malate (the flavin/FAD active site is that of the MQO/DAO glutathione-reductase-2-like FAD superfamily), and its "substrate" on the acceptor side is a respiratory quinone (ubiquinone in aerobically grown *Pseudomonas*).

The identity of the target is confirmed: gene symbol, EC number, MQO family (Pfam PF06039; InterPro IPR006231 "MQO", IPR036188 "FAD/NAD(P)-binding domain superfamily"), organism (*P. putida* KT2440, taxid 160488), and locus tag (PP_2925) are all mutually consistent across UniProt, KEGG, and the primary MQO literature. There is no ambiguity of identity; the caveat is

only that no experimental study has characterized **this specific paralog** — its function is inferred from the well-characterized bacterial MQO family plus bioinformatic, structural, and network evidence.

2. Gene/Protein Identity and Verification

Attribute	Value	Source
Gene name / synonym	<i>mqo3</i> / <i>mgo-3</i> / <i>mgo-III</i>	UniProt Q88IS4; KEGG ppu:PP_2925
Locus tag	PP_2925	UniProt; KEGG
Protein	Probable malate:quinone oxidoreductase 3 (MQO 3)	UniProt (HAMAP-Rule MF_00212)
EC	1.1.5.4 (malate dehydrogenase, quinone)	UniProt; KEGG
Length	539 aa	UniProt
Cofactor	FAD (non-covalent)	UniProt; Molenaar 1998 (P 9660197)
Family/domains	MQO family; Pfam PF06039 (Mgo), also DAO; IPR006231, IPR036188	UniProt/InterPro; KEGG MOTIF
Organism	<i>Pseudomonas putida</i> KT2440 (taxid 160488)	UniProt
Reference proteome	UP000000556	UniProt

All identifiers are internally consistent, and the retrieved primary literature (below) is genuinely about **malate:quinone oxidoreductase**, not a same-symbol homonym. Verification passed.

3. Primary Function: The Catalyzed Reaction and Substrate Specificity

MQO is a **flavin adenine dinucleotide (FAD)-dependent oxidoreductase** that oxidizes L-malate to oxaloacetate and reduces a quinone. Unlike the classical NAD-dependent malate dehydrogenase (MDH, EC 1.1.1.37), MQO **does not use NAD⁺**; the natural electron acceptor is a membrane quinone. This was established biochemically for the founding family member from *Corynebacterium glutamicum*:

"It oxidizes L-malate to oxaloacetate and donates electrons to ubiquinone-1 and other artificial acceptors or, via the electron transfer chain, to oxygen. NAD is not an acceptor and the natural direct acceptor for the enzyme is most likely a quinone." (Molenaar et al., 1998,

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The same study established that **FAD is the (non-covalently but tightly bound) prosthetic group** and that the enzyme is lipid-activated (Molenaar et al., 1998,

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MQO3's sequence contains the canonical N-proximal FAD/dinucleotide-binding fingerprint (VDVLLVGGGIMS), a Rossmann-like $\beta\alpha\beta$ motif of the FAD/NAD(P)-binding superfamily (IPR036188), shared by all three *P. putida* paralogs.

Thermodynamic/directional consequence. Because MQO deposits electrons into the quinone pool (high redox potential relative to NAD⁺) rather than reducing NAD⁺, the malate→oxaloacetate reaction becomes **effectively irreversible in the oxidative direction**, in contrast to the readily reversible NAD-MDH. This "pulls" the TCA cycle forward and directly links carbon oxidation to respiratory electron flow. The purified thermophilic *Bacillus* PS3 MQO directly demonstrated menaquinol production that could feed a quinol oxidase, confirming quinone reduction as the physiological output (Kabashima et al., 2013,

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Substrate specificity. The carbon substrate is L-(S)-malate; the acceptor is a respiratory quinone (ubiquinone in aerobic *Pseudomonas*; menaquinone in organisms that use it). This is an inference from the conserved MQO family fold and the UniProt/Rhea catalytic annotation; no paralog-specific kinetic study exists for PP_2925.

4. Biological Process / Pathway Context

MQO3 operates in **central carbon metabolism**, specifically the **TCA cycle** (KEGG ppu00020; module M00009), and is mapped by KEGG additionally to **pyruvate metabolism** (ppu00620), the **methylcitrate cycle** (M00982), and general **carbon metabolism** (ppu01200). Its reaction sits at the **malate ↔ oxaloacetate node**, a major hub connecting:

- the **oxidative TCA cycle** (fumarate → malate → oxaloacetate → citrate),
- the **glyoxylate shunt** (isocitrate lyase/malate synthase; important during growth on acetyl-CoA-generating substrates), and
- **anaplerosis/gluconeogenesis** (oxaloacetate ↔ PEP/pyruvate via PEP carboxylase, pyruvate carboxylase, malic enzyme, PEP carboxykinase).

This placement is independently confirmed by the **STRING functional-association network** for PP_2925, whose highest-scoring partners are precisely the neighbours of this node: fumarate hydratase (fumC-I/fumC-II, PP_0897; produces MQO's substrate malate), **citrate synthase gltA** (consumes MQO's product oxaloacetate), **malate synthase glcB** (glyoxylate-shunt malate producer), the parallel **NAD-MDH (mdh)**, and the anaplerotic enzymes **PEP carboxylase (ppc)**, **pyruvate carboxylase (pycB)**, and **malic enzyme (maeB)** — together with the two sister paralogs mqo-I and mqo-II (all association scores 0.90–0.94).

Functional relevance in *P. putida* physiology: *P. putida* is an obligately aerobic, metabolically versatile soil bacterium with a strongly **oxidative, respiration-driven metabolism** and a highly active TCA cycle that is central to its catabolism of organic acids and of **aromatic/lignin-derived compounds** (which funnel through acetyl-CoA and the glyoxylate/TCA cycle) (e.g., Zhou et al., 2025,

Moreno et al., 2024,

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A respiration-coupled malate oxidase such as MQO fits this lifestyle by maximizing the yield of reducing equivalents delivered to the electron-transport chain.

In the related organism *P. aeruginosa*, **mqo is required for growth on ethanol** (which enters metabolism as acetyl-CoA and depends on a functional glyoxylate/TCA cycle to regenerate oxaloacetate) (Görisch, 2003,

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— a concrete example of the physiological importance of MQO-type malate oxidation in pseudomonads.

5. Subcellular Localization

MQO family enzymes are **peripheral membrane proteins** associated with the cytoplasmic (inner) membrane, from which they can be released by chelators:

"Mqo is a peripheral membrane protein and can be released from the membrane by addition of chelators." (Molenaar et al., 1998,

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They are membrane-associated (not integral polytopic membrane proteins) and function at the **cytoplasmic face of the inner membrane**, where the catalytic FAD domain oxidizes cytosolic malate and reduces the membrane quinone pool. In *H. pylori* MQO activity was demonstrated directly in isolated membranes, and the enzyme was described as being — "similar to succinate dehydrogenase" — a shared component of **both the electron transfer chain and the citric acid cycle** (Kather et al., 2000,

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By homology, MQO3 of *P. putida* is expected to localize to the inner membrane and reduce ubiquinone.

Paralog-specific feature. Uniquely among the three *P. putida* MQOs, MQO3 carries an **N-terminal ~24-residue extension** (MKITIKPKLAMAGLALAIGMLNAQAA...) that has the classic tripartite architecture of a **Sec-type signal peptide** (positively charged n-region KITIKPK; hydrophobic h-region LAMAGLALAIGML; an A-x-A signal-peptidase-I cleavage motif ...AQAA↓ET), after which the sequence aligns residue-for-residue with the N-terminus of MQO2. The AlphaFold model (AF-Q88IS4; mean pLDDT 88.8) shows these first ~24 residues are **disordered/very-low-confidence (per-residue pLDDT ~37–49)** and structurally separate from the well-folded FAD-catalytic core (confidence rises sharply at residues ~25–30). This is consistent with a **flexible, membrane-targeting or cleavable N-terminal add-on** distinct from the catalytic domain. *Caveat:* canonical MQOs are cytoplasm-facing peripheral membrane enzymes (they must access the cytoplasmic quinone pool), so genuine periplasmic export is unlikely; the extension is more plausibly a membrane-anchoring/targeting or regulatory segment. UniProt/HAMAP annotate no signal-peptide feature. This should be treated as a bioinformatic hypothesis pending experimental localization.

6. Evolutionary / Paralog Context

P. putida KT2440 encodes **three MQO paralogs** — mqo1/PP_0751 (502 aa), mqo2/PP_1251 (527 aa), mqo3/PP_2925 (539 aa), all assigned KEGG orthology K00116 / EC 1.1.5.4 — **plus a single canonical NAD-dependent MDH** (mdh/PP_0654; K00024; EC 1.1.1.37). Global pairwise alignment shows **mqo3 and mqo2 are recent, close paralogs (78.0% identity)**, whereas mqo1 is more anciently diverged (mqo3–mqo1 50.7%; mqo2–mqo1 48.2%). Thus mqo3 most likely arose by a relatively recent duplication of an mqo2-like ancestor, with the acquisition/retention of its distinctive N-terminal extension as the main sequence divergence. Expansion of MQO gene families is characteristic of metabolically versatile *Pseudomonas*.

The presence of both MQO and NAD-MDH routes implies **functional redundancy** at the malate-oxidation step. In *E. coli*, MQO and MDH are active simultaneously, but Δmqo has no measurable growth or fitness defect while Δmdh is severely impaired — i.e., MQO is dispensable when MDH is present (van der Rest et al., 2000,).

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By analogy, MQO3 is expected to be **conditionally used and non-essential** in *P. putida*, rather than the principal malate-oxidizing enzyme, especially given the additional redundancy from mqo1 and mqo2.

7. Regulation (family-level inference)

MQO expression is tied to respiratory/redox status. In *E. coli*, *mgo* transcription is controlled by the **ArcA-ArcB redox two-component system**, and MQO activity peaks during exponential (respiratory) growth and falls at stationary phase (van der Rest et al., 2000,).

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MQO activity is also regulated by carbon/energy source. In *C. glutamicum*, loss of MQO down-regulates TCA-cycle gene expression and **redirects oxaloacetate flux** (e.g., toward L-lysine biosynthesis), illustrating that MQO activity influences the partitioning of the oxaloacetate node (Mitsuhashi et al., 2006,).

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Paralog-specific regulation of PP_2925 in *P. putida* has not been experimentally defined.

8. Supported and Refuted Hypotheses

Supported: - H1 (primary function): MQO3 catalyzes FAD-dependent oxidation of L-malate to oxaloacetate with a quinone acceptor (EC 1.1.5.4). *Supported* by UniProt/Rhea catalytic annotation, conserved FAD-binding fingerprint, MQO-family membership, and biochemically characterized homologs

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- **H2 (localization):** MQO3 is a peripheral, inner-membrane-associated flavoprotein acting at the cytoplasmic face. *Supported* by family biochemistry

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- **H3 (pathway):** MQO3 sits at the malate↔oxaloacetate node of the TCA/glyoxylate/anaplerotic network, coupling the cycle to respiration. *Supported* by KEGG pathway mapping and STRING network (partners: fumarase, citrate synthase, malate synthase, MDH, Ppc/PycB/MaeB). - **H4 (evolution/redundancy):** *mgo3* is a recent duplicate of *mgo2* and functionally redundant with

NAD-MDH and the other MQOs. *Supported* by pairwise identity analysis and the *E. coli* $\Delta m q o$ precedent

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Refuted / down-weighted: - H5 (periplasmic export via the N-terminal signal): Although MQO3's N-terminal extension has Sec-signal-peptide architecture and is disordered in the AlphaFold model, genuine periplasmic localization is **inconsistent** with the requirement for MQO to reduce the cytoplasmic quinone pool. This hypothesis is down-weighted in favour of a membrane-targeting/regulatory role; it remains untested experimentally.

9. Evidence Quality and Limitations

- **Strengths:** The core annotation (reaction, cofactor, localization, pathway) rests on decades of biochemical and genetic characterization of the MQO family in *C. glutamicum*, *E. coli*, *H. pylori*, and *Bacillus*, plus fully consistent, curated database records (UniProt HAMAP MF_00212, KEGG, Rhea, InterPro) and independent structural (AlphaFold) and network (STRING) corroboration.
- **Evidence level:** UniProt classifies all three *P. putida* MQO paralogs (Q88IS4/mqo3, Q88NF9/mqo2, Q88PU7/mqo1) at protein-existence level "**3: Inferred from homology**", with no proteomics cross-references — consistent with the "Probable" designation and the absence of paralog-specific experimental work.
- **Limitations:** No experimental study has directly characterized **PP_2925/MQO3 itself** — kinetics, quinone specificity, exact subcellular topology, the role of the N-terminal extension, and the physiological division of labour among mqo1/mqo2/mqo3 in *P. putida* are all inferred, not measured. The annotation status "Probable" reflects this. High-throughput RB-TnSeq fitness datasets exist for *P. putida* (e.g.,

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and could, in principle, quantify MQO3's conditional fitness contribution, but paralog-specific values were not extracted here.

10. Future Directions

1. Targeted deletion of PP_2925 (and combinatorial $\Delta m q o 1/2/3$ and $\Delta m d h$ mutants) to define the division of labour and essentiality at the malate-oxidation step.
 2. Enzymatic characterization of purified MQO3 (kinetics, ubiquinone vs. menaquinone specificity, FAD stoichiometry).
 3. Experimental determination of subcellular topology and the fate/function of the N-terminal extension (cleavage, membrane anchoring, or regulation).
 4. Condition-resolved proteomics to determine which paralog is expressed on which carbon source (organic acids vs. sugars vs. aromatics).
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Key References

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